ch20 Electrostatics

2012 Q24

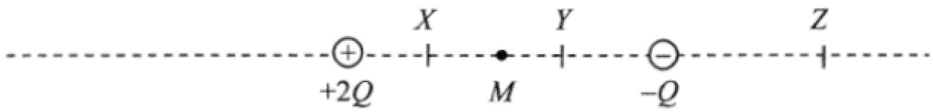
24. P, Q, R, S are charged objects. When two of them are brought close to each other, P and Q repel, R and S also repel while Q and R attract each other. Which of the following descriptions about their charges is/are possible ?

(1) P and R are negatively charged.
(2) Q and S are positively charged.
(3) P is positively charged and S is negatively charged.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2012 Q25

*25.



Two point charges $+2Q$ and $-Q$ are situated at fixed positions as shown. M is the mid-point between the charges. X, Y and Z are points marked on the line joining these two charges. At which point could

(1) the resultant electric field due to the two charges be zero ?
(2) the total electric potential due to the two charges be zero ?

	(1)	(2)
A.	Z	X
B.	Z	Y
C.	X	Z
D.	Y	Z

2013 Q24

24. X and Y are two small identical metal spheres carrying charges $-2Q$ and $+6Q$ respectively. When X and Y are separated by a certain distance, the magnitude of the electrostatic force between them is F.

The spheres are brought to touch each other and then placed back to their original positions. The electrostatic force between them becomes

A. $\frac{1}{4}F$, attractive.

B. $\frac{1}{4}F$, repulsive.

C. $\frac{1}{3}F$, attractive.

D. $\frac{1}{3}F$, repulsive.

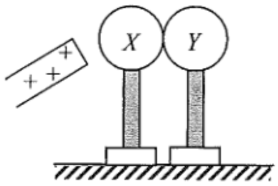
2013 Q25

*25. Lightning flash may occur when the strength of the electric field (assumed uniform) between a thundercloud and the ground reaches 3×10^6 N C^{-1}. A lightning flash on average discharges about 20 C of charge. If a thundercloud is at a height of 500 m above the ground, estimate the order of magnitude of the energy released in a lightning flash.

A. 10^6 J
B. 10^8 J
C. 10^{10} J
D. 10^{12} J

2014 Q20

20.

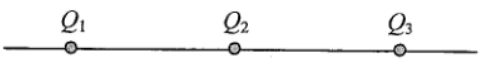


Two insulated uncharged metal spheres X and Y are placed in contact. A positively-charged rod is brought near X as shown. X is then touched by a finger momentarily and the two spheres are then separated by removing Y. The charged rod is removed afterwards. Which of the following describes the charges on X and Y?

	sphere X	sphere Y
A.	uncharged	uncharged
B.	uncharged	positive
C.	negative	uncharged
D.	negative	negative

2014 Q21

21.

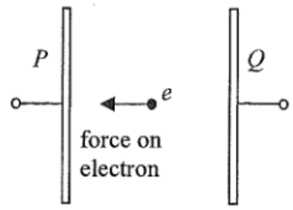


Three point charges Q_1, Q_2 and Q_3 are fixed on a straight line with Q_2 at the mid-point of Q_1 and Q_3. The resultant electrostatic force on each charge is zero. Which of the following can be the sign and magnitude (in the same arbitrary units) of Q_1, Q_2 and Q_3?

	Q_1	Q_2	Q_3
A.	+2	+1	+2
B.	+2	−1	+2
C.	−4	+1	+4
D.	−4	+1	−4

2014 Q22

22.

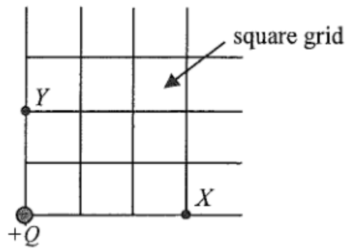


Two parallel metal plates P and Q are maintained at a certain p.d. by a battery (not shown in the figure). An electron placed between the plates would experience an electrostatic force of 8.0×10^{-18} N towards P. Which of the following descriptions about the electric field E between the plates is correct ?

A. $E = 0.02\ \mathrm{N\,C^{-1}}$ from Q to P.
B. $E = 0.02\ \mathrm{N\,C^{-1}}$ from P to Q.
C. $E = 50\ \mathrm{N\,C^{-1}}$ from Q to P.
D. $E = 50\ \mathrm{N\,C^{-1}}$ from P to Q.

2014 Q23

*23.



The figure shows the location of an isolated point charge $+Q$. If the electric potential at X is V, what is the electric potential at Y ?

A. $\frac{2}{\text{[illegible]}}V$

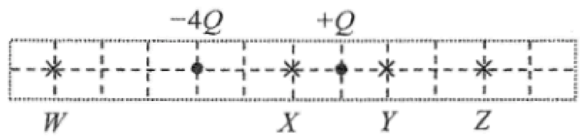
2015 Q21

21. Three conducting balls are suspended by insulating threads. **Any two of them** are found to attract each other if brought close to each other. Which conclusion below is correct ?

A. Only one ball is uncharged while the other two carry like charges.
B. Only one ball is uncharged while the other two carry unlike charges.
C. Only one ball is charged.
D. All three balls are charged.

2015 Q22

22.



Two point charges $-4Q$ and $+Q$ are fixed as shown. At which point indicated in the figure is the resultant electric field due to these two charges zero ?

A. W
B. X
C. Y
D. Z

2015 Q23

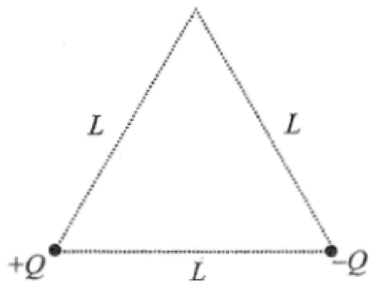
*23. Point charges $+Q$ and $-Q$ are fixed respectively at two vertices of an equilateral triangle with sides of length L as shown. What is the minimum energy required for bringing another point charge $+Q$ from infinity to the third vertex ?

A. 0

B. $\dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{Q^2}{L}\right)$

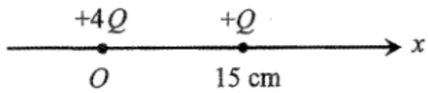
C. $\dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{2Q^2}{L}\right)$

D. $\dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{3Q^2}{L}\right)$



2016 Q24

*24.

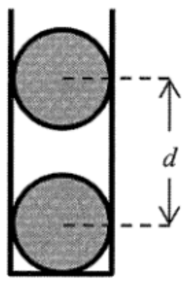


Point charges $+4Q$ and $+Q$ are fixed on the x-axis with $+4Q$ at the origin O and $+Q$ at x = 15 cm as shown. The respective electric fields due to the two charges are equal at

A. x = 10 cm.
B. x = 12 cm.
C. x = 20 cm.
D. x = 30 cm.

2017 Q22

22. In the figure, two charged conducting spheres of the same mass m are put in a vertical plastic cylinder. The inner wall of the cylinder is smooth. The spheres are separated by a distance d and remain in equilibrium.

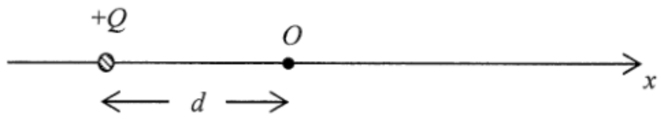


Which of the following statements **MUST BE** correct ?

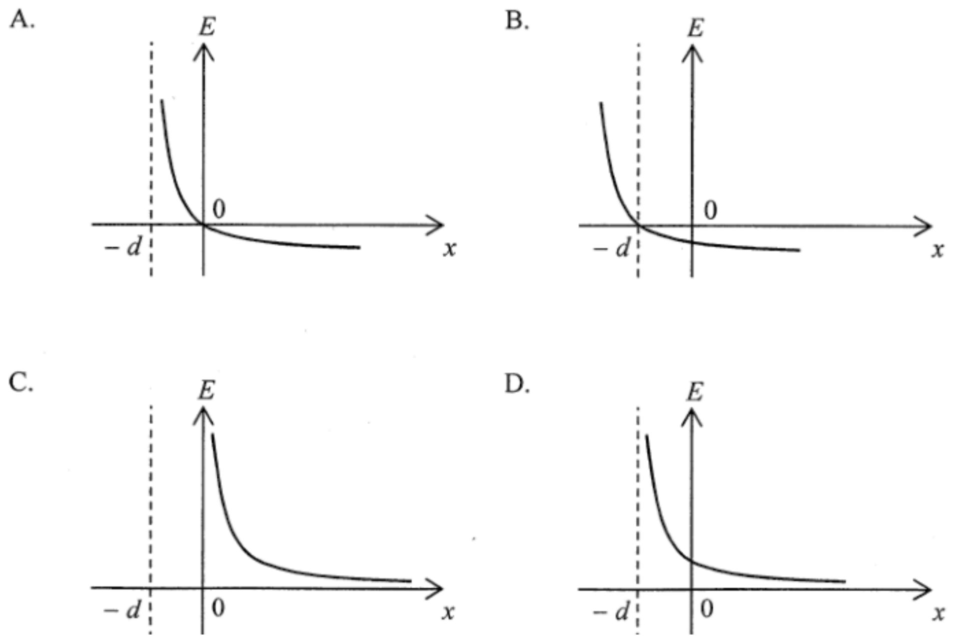
(1) Both spheres carry positive charges.
(2) The amount of charges on the two spheres is the same.
(3) The separation d depends on m.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

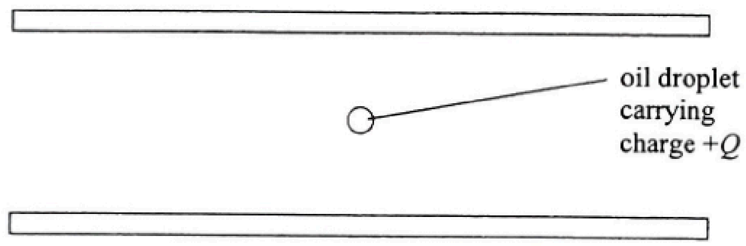
*23. A point charge $+Q$ is fixed at a distance d away from the origin O as shown.



Which of the following graphs best represents the variation of the electric field strength E along the x-axis ? (Take the electric field pointing to the right as positive.)



22.



A charged oil droplet of mass m is found floating between two horizontal parallel metal plates with an electric field of constant strength in between. The oil droplet is carrying charge $+Q$. What are the direction and strength of the electric field ?

	direction	strength
A.	upward	$\frac{mg}{Q}$
B.	upward	$\frac{Q}{mg}$
C.	downward	$\frac{mg}{Q}$
D.	downward	$\frac{Q}{mg}$

2019 Q23

23.

In the above figure, point charge Y is placed in the middle of two identical positive point charges X and Z, with Z being fixed. Both X and Y are in equilibrium and at rest initially. What would happen to X if Y is slightly pushed towards Z ?

A. It moves towards the left.
B. It moves towards the right.
C. It remains at rest.
D. It cannot be determined as the sign of Y is not known.

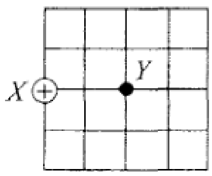
2021 Q21

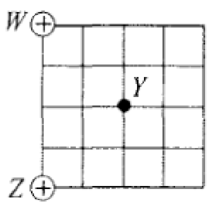
21. Three isolated identical metal spheres X, Y and Z carry charges $+2Q$, $-4Q$ and $+5Q$ respectively. Y is first moved to touch X and then Y is brought in contact with Z. When Y and Z are separated, find the charge on each sphere.

	X	Y	Z
A.	0	$+1.5Q$	$+1.5Q$
B.	$-Q$	$+0.5Q$	$+0.5Q$
C.	$+Q$	$+Q$	$+Q$
D.	$-Q$	$+2Q$	$+2Q$

2021 Q22

*22. When a point charge $+Q$ is placed at X as shown, the strength of the electric field at Y is E_0.



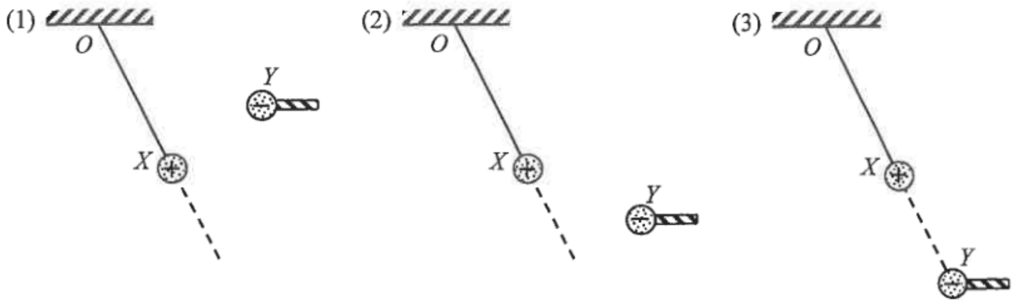


If W and Z are each placed with a point charge of $+Q$, what will be the electric field strength at Y ?
Note: $\sin 45° = \cos 45° = \frac{\sqrt{2}}{2}$

A. $\frac{\sqrt{2}}{2} E_0$
B. E_0
C. $\sqrt{2}\, E_0$
D. $2\, E_0$

2022 Q23

23. A positively charged sphere X of mass m is suspended from a fixed point O by a nylon thread. Another negatively charged sphere Y at the end of an insulating rod is held in various positions as shown. O, X and Y are in the same vertical plane.

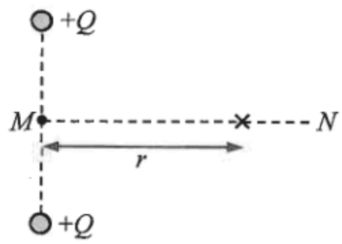


In which situation(s) could X be kept at rest as shown ?

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

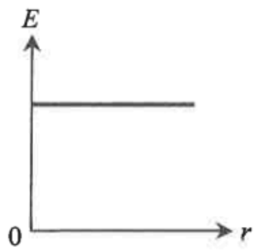
2022 Q24

*24.

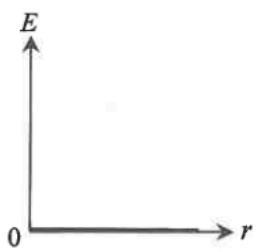


Two positive point charges $+Q$ are fixed as shown above. MN is the perpendicular bisector of the line joining the charges. Which graph correctly shows the variation of the electric field strength E on line MN with distance r from M ?

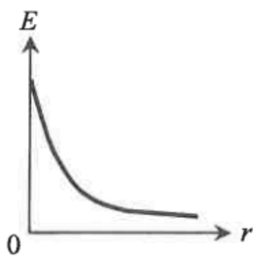
A.



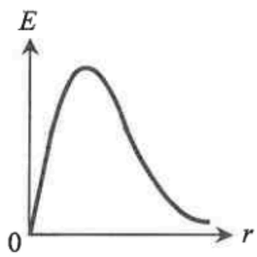
B.



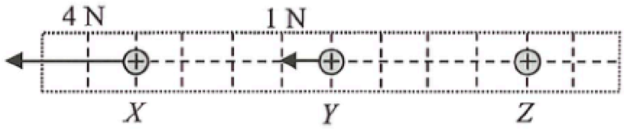
C.



D.



23. Positive point charges X, Y and Z are placed along a straight line with Y at the mid-point between X and Z. Assume that the only interaction among them is the electrostatic force.

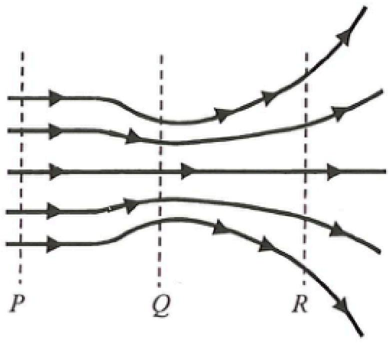


At the instant shown, the net forces acting on X and Y respectively are 4 N and 1 N both towards the left. At that instant, which statements about Z are correct ?

(1) The net force acting on Z is towards the right.
(2) The net force acting on Z is 5 N in magnitude.
(3) The charge of Z must be larger than that of X.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

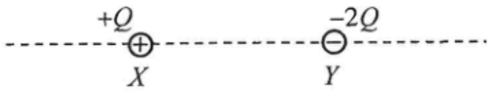
24. The figure shows the field lines of an electric field. Arrange the respective electric field strength E_P, E_Q and E_R of regions P, Q and R in ascending order.



A. $E_P < E_Q < E_R$
B. $E_Q < E_R < E_P$
C. $E_R < E_P < E_Q$
D. $E_Q < E_P < E_R$

2024 Q23

23. Charges $+Q$ and $-2Q$ are fixed at X and Y as shown.



Which description(s) about the electric field along the dotted line is/are correct ?

(1) Between X and Y, the respective electric fields due to the two charges are in the same direction.
(2) To the right of Y, the resultant electric field is always directed to the left.
(3) On the dotted line, there exist two points where the resultant electric field is zero.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

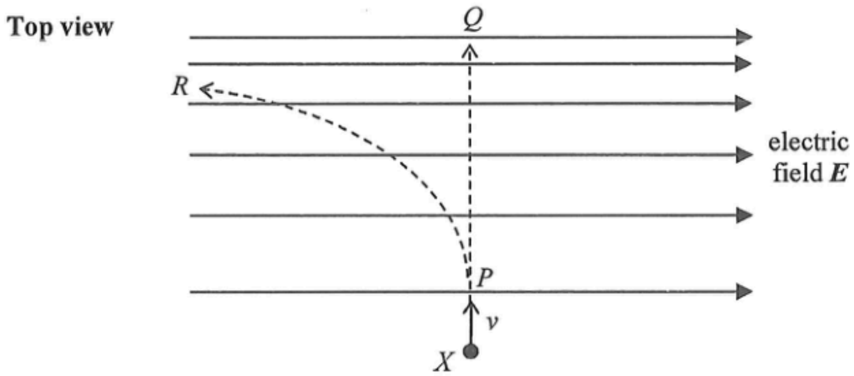
2025 Q22

22. Three light conducting spheres are suspended by nylon threads. Any two of them are found to attract each other when they are brought close to each other. Which conclusion below is correct ?

A. All three spheres are charged.
B. Only one sphere is charged.
C. Only one sphere is uncharged and the other two carry like charges.
D. Only one sphere is uncharged and the other two carry unlike charges.

2025 Q23

23. A negatively charged particle X enters an electric field with initial velocity v perpendicular to the field lines as shown. Neglect the effects of gravity.

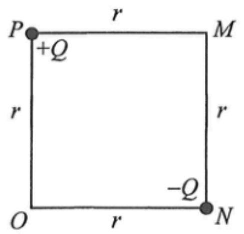


Particle X may move along

A. PQ with uniform speed.
B. PQ with increasing speed.
C. PR with decreasing speed.
D. PR with increasing speed.

2025 Q24

*24.



Point charges $+Q$ and $-Q$ are fixed respectively at corners P and N of square $MNOP$ of side length r as shown above. What is the electric field at O ?

A. $\frac{\sqrt{2}Q}{4\pi\varepsilon_0 r^2}$ along the direction from P to N

B. $\frac{Q}{4\pi\varepsilon_0 r^2}$ along the direction from P to N

C. $\frac{\sqrt{2}Q}{4\pi\varepsilon_0 r^2}$ along the direction from N to P

D. $\frac{Q}{4\pi\varepsilon_0 r^2}$ along the direction from N to P

2026 Q21

21. Two small charged spheres exert an electrostatic force of magnitude F on each other. If the amount of charge on each sphere is doubled and the distance between them is also doubled, the magnitude of the electrostatic force becomes

A. $\frac{F}{2}$.

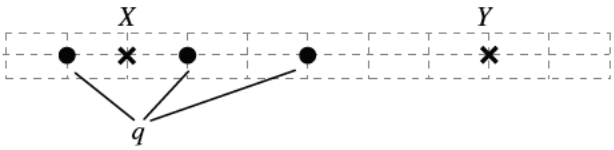
B. F.

C. $2F$.

D. $4F$.

pp Q24

24.

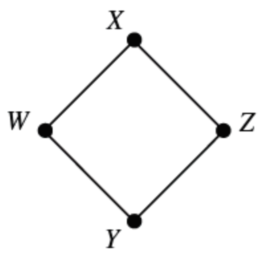


Three identical point charges q (represented by dots) are situated in the space as shown. Which of the following descriptions about the direction and magnitude of the electric field E at X and at Y is correct ?

	Direction	Magnitude
A.	Same	$E_X > E_Y$
B.	Same	$E_X < E_Y$
C.	Opposite	$E_X > E_Y$
D.	Opposite	$E_X < E_Y$

pp Q25

*25.



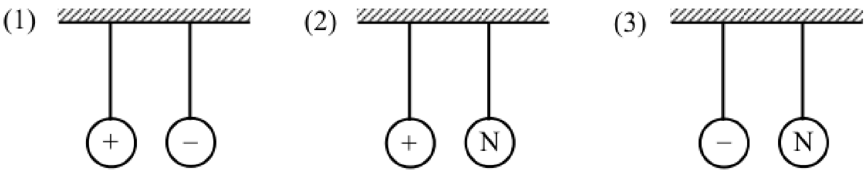
The figure above shows four points W, X, Y and Z in a uniform electric field. $WXZY$ is a square. The electric potential at W, X and Y are 1 V, 5 V and 5 V respectively. What is the electric potential at Z ?

A. 1 V
B. 6 V
C. 9 V
D. 11 V

sap Q24

24. Two conducting spheres are hanging freely in air by insulating threads. In which of the following will the two spheres attract each other ?

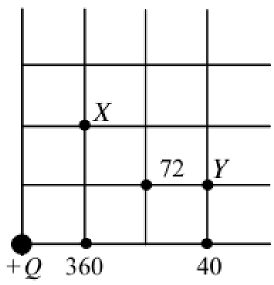
Note : 'N' denotes that the sphere is uncharged.



A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

sap Q32

32.



The figure shows the location of an isolated charge of size $+Q$. The size (in an arbitrary unit) of the electric field strength is marked at certain points. What is the size (in the same arbitrary unit) of the electric field strength at X and Y ?

	electric field strength at *X*	**electric field strength at *Y***
A.	72	30
B.	72	36
C.	90	30
D.	90	36